



Detecting Meaningful Locations and Usual Environment

Module 2 - Session 2.4.2

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Agenda

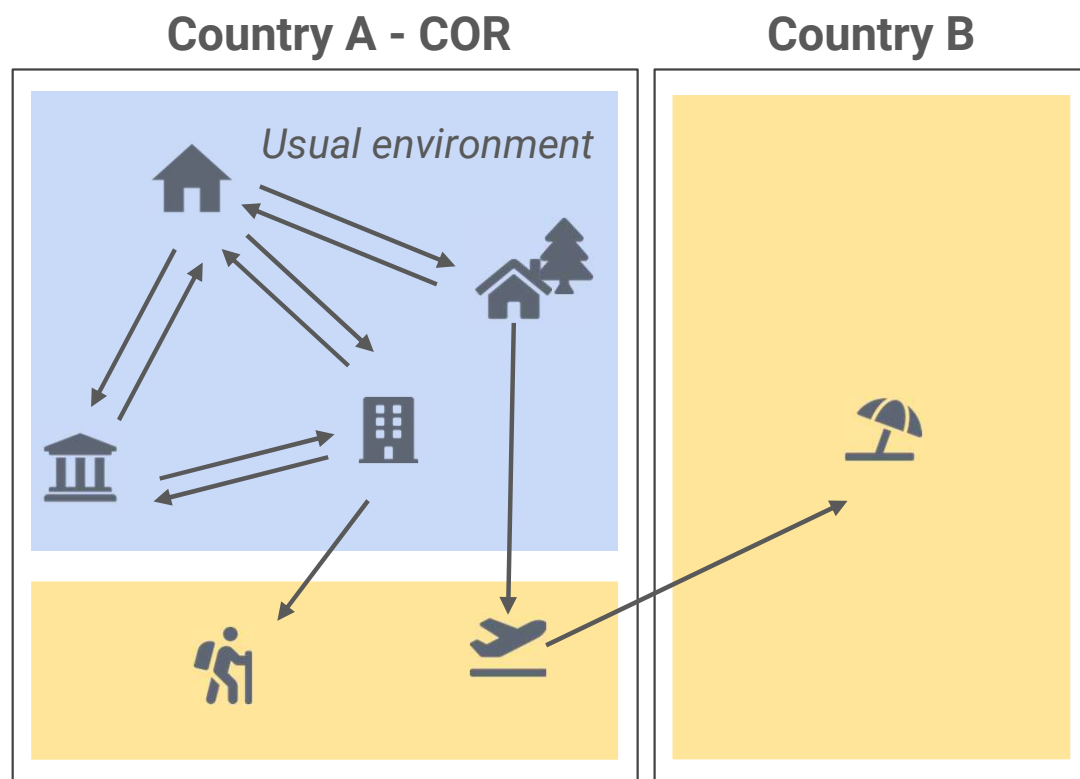
- Definitions - meaningful locations, usual environment, country of residence;
- Data requirements for meaningful location detection;
- Detecting anchor points as meaningful locations from mobile positioning data;
- Model validation;
- Defining usual environment;
- Identifying country of residence;
- Limitations and ethical considerations.

Meaningful locations, usual environment and country of residence

Meaningful locations - *regularly* visited places that have meaning for individuals. Examples can include home, work, school. Also referred to as **anchor points**.

Usual environment (UE) - the **geographical boundaries** within which subscribers or individuals displace themselves when going about their **daily activities**. It can be defined with the precision of country of reference, region, municipality or some other geographical areas or administrative units.

Country of residence (COR) - A person's country of **usual residence** is that in which the person lives, that is to say, the country in which the person has a place to live where he or she normally spends the daily period of rest.



Why do we need to determine meaningful locations, country of residence and usual environment?

The concepts of *meaningful locations*, *country of residence* and *usual environment* are the cornerstones of MPD methodology that enable various different use cases requiring the understanding of home location, residentship, and travel patterns between different places.



Official statistics



Transportation planning



Urban planning



Crisis and disaster management



Tourism statistics



Epidemiology



Public transportation planning

Using MPD for detecting meaningful locations

- Mobile phone data (MPD) enables to create a snapshot of **temporal** and **spatial patterns of phone users**.
- The attributes MPD typically offers are **user related identities**, **timestamp of the event** and **location for the event**.
- Therefore, detecting location semantics from MPD typically focuses on **identifying characteristics** related to usage of a single cell or cluster of cells per defined time period (day, week, month, year) per phone subscriber.

DOMESTIC	OUTBOUND	INBOUND
• ID	• ID	• ID ID for a person
• TIMESTAMP	• TIMESTAMP	• TIMESTAMP
• CELL_ID	• COUNTRY	• CELL_ID
	• CELL_ID*	• COUNTRY Location (MCC from IMSI)

Data requirements for meaningful location detection

Is the location of a mobile call something special, or it is just another location of another activity?²

At **minimum**, MPD needs to allow:

- detection of visitation **regularity** and **frequency** over time;
- **subscriber identity consistency** - detection of the same person over period of time.

Requirements for MPD **temporal coverage** can be **use case dependent**. Some examples:

- understanding commuting patterns - shorter time period could already provide useful insights;
- detecting MPD-based place of residence for registry based census - official definition for place of residence contains the clause that there exists intention to stay for a year or more.

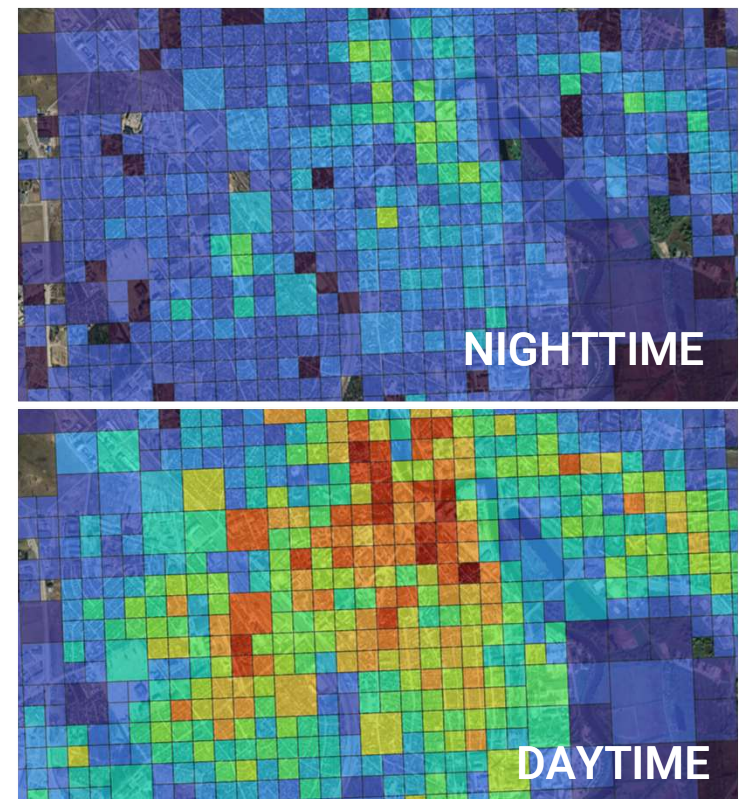
²Ahas, R et al (2010) *Using Mobile Positioning Data to Model Locations Meaningful to Users of Mobile Phones*

Meaningful locations in MPD

Anchor points - meaningful locations detected from MPD data.

Types of anchor points typically detected from MPD:

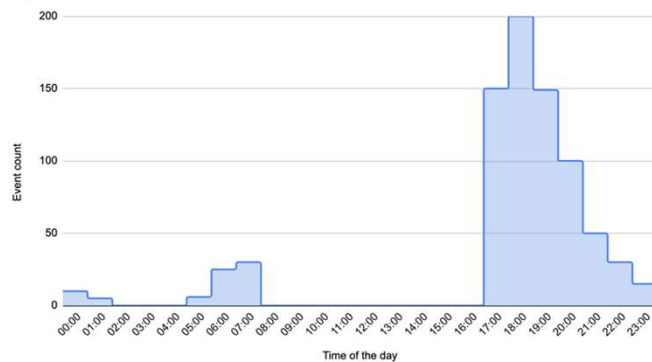
- home anchor;
- work-time anchor;
- multifunctional anchor;
- secondary home(s) anchor(s);
- secondary work-time anchor(s);
- other regularly visited places.



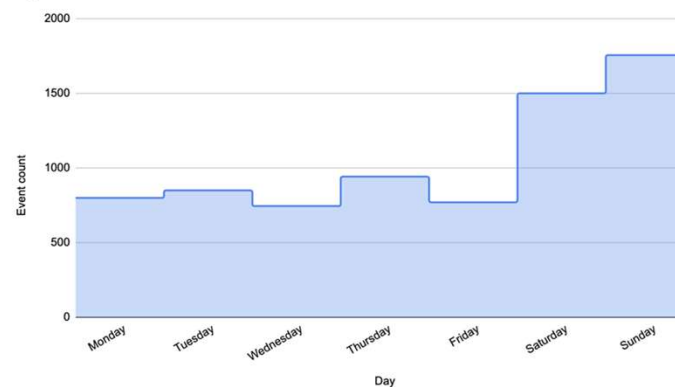
Anchor point detection: home location

- Detecting the **place of residence** is arguably the most important first step in building a model to produce various statistics. For example, it acts as input for:
 - population, domestic tourism, internal migration, de facto population, mobility statistics.
- **Choice of home detection algorithm** influences the result and all statistics that follow.
- What **characterises** a **home**? **How long** do I need to **observe a location** to detect a home? How do we infer **semantics** for a location?

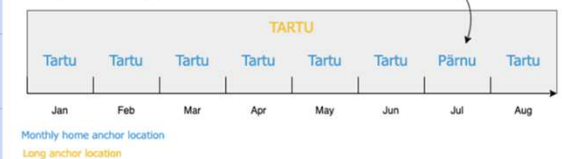
Typical weekday: home location



Typical week: home location



Prominence



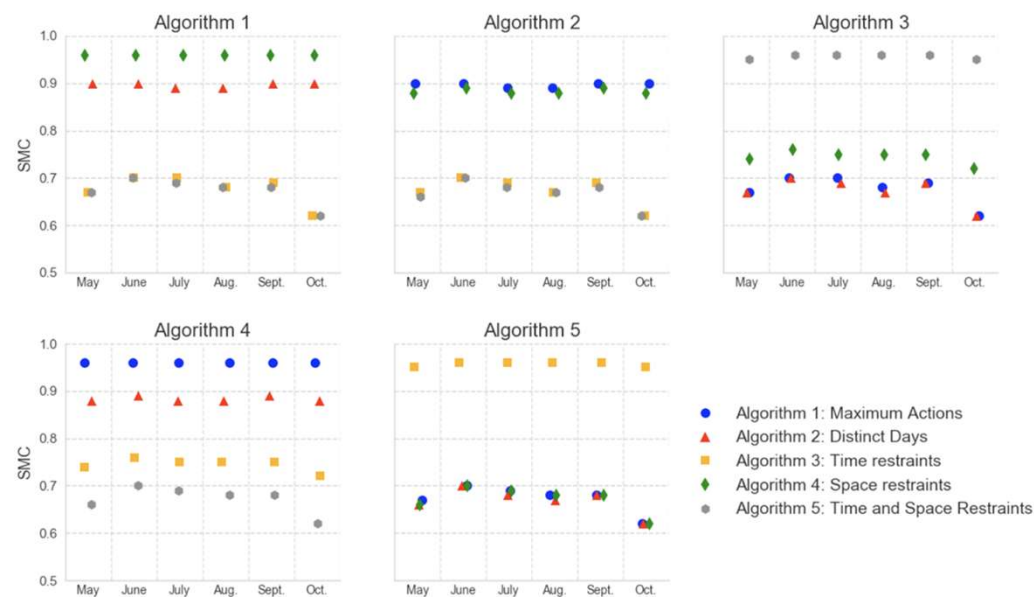
Anchor point detection: home location

Method	Temporal variables	Activity variables	Example
<i>Heuristic</i> and <i>statistical</i> methods in tree-based <i>decision rules</i>	<i>Day of the week</i> , generalized <i>day</i> - and <i>nighttime</i>	<i>Frequency</i> of activity, <i>distinct days</i> of activity, <i>standard deviation</i> of activity start times	Ahas, R et al (2010) <i>Using Mobile Positioning Data to Model Locations Meaningful to Users of Mobile Phones</i>
<i>Heuristic</i>	Generalized <i>nighttime</i> window	<i>Frequency</i> of activity	Calabrese, F et al (2013) <i>Understanding individual mobility patterns from urban sensing data: A mobile phone trace example</i>
			Phithakkitnukoon, S et al (2012) <i>Socio-Geography of Human Mobility: A Study Using Longitudinal Mobile Phone Data</i>
<i>Heuristic</i> methods, <i>clustering</i> , <i>decision rules</i>	Individualized <i>resting</i> and <i>working</i> hours	<i>Frequency</i> of activity, activity <i>location</i> (land use)	Yang, Y (2021) <i>Detecting Home and Work Locations from Mobile Phone Cellular Signaling Data</i>
<i>Heuristic</i> and <i>statistical</i> methods, <i>clustering</i>	Generalized <i>resting</i> and <i>working</i> hours, weekdays and weekends	<i>Frequency</i> of activity	Isaacman, S et al (2011) <i>Identifying Important Places in People's Lives from Cellular Network Data</i>
<i>Heuristic</i> methods	Generalized <i>resting</i> and <i>working</i> hours, <i>weekdays</i> and <i>weekends</i>	<i>Duration</i> of stay	N, A R S et al (2021) <i>Home and Work Identification Process Using Mobile Positioning Data</i>

Anchor point detection: algorithm selection

- Validation study¹ compared different home detection algorithms to census counts at cell tower level in France.
- The findings show that the choice of criteria in home detection algorithms influences the results significantly (up to 40%).

¹Vanhoof, M et al (2018) Assessing the quality of home detection from mobile phone data for official statistics



SMC values express the ratio of users to whom two HDAs detect the same home.

Anchor point detection: work-time location

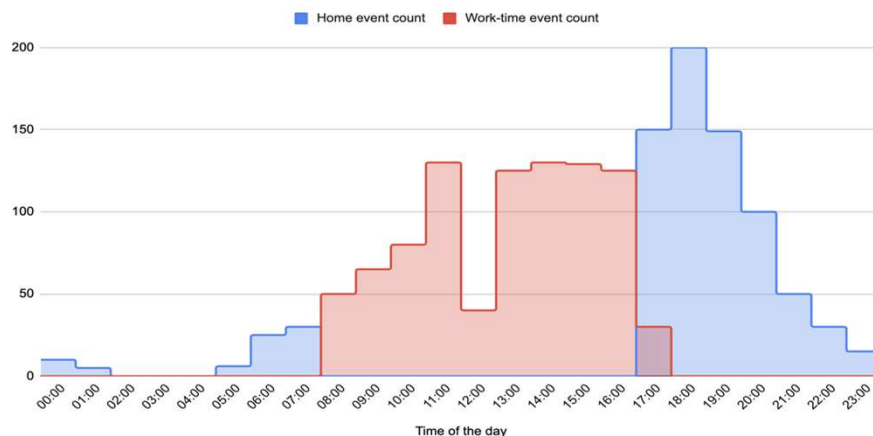
- **Work-time** vs **work** location: MPD typically allows detection of work-time location as we cannot infer from the data facts about the employment.
- Literature doesn't always make a distinction here, but **conceptually** it should be understood - relevant for **result interpretation**.

Work-time location	Work-location
Location of a mobile device during pre-defined working hours. An activity and its location that falls into the working hours window , e.g. locations for school, university and work related activities.	Location where the user performs their employment activities. An activity and its location strictly related to employment .

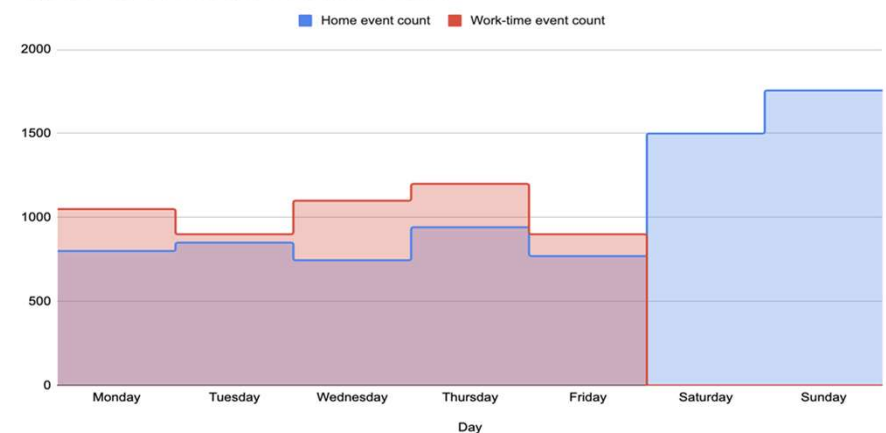
Anchor point detection: work-time location

- **Work-time location detection** acts as an important input for several MPD-based use cases. For example:
 - mobility statistics (e.g. commuting OD-matrix), de facto population (who are present at a location?), tourism (identifying usual environment).
- What **characterises** work-time location? How to account for diverse profile of people (e.g. nighttime workers, people who move around a lot)?

Typical weekday: home and work-time location



Typical week: home and work-time location



Anchor point detection: work-time location

Method	Temporal variables	Activity variables	Diverse work profiles	Example
<i>Heuristic and statistical methods</i> in tree-based <i>decision rules</i>	<i>Start time of calls, standard deviation</i> of daily event starting times, <i>day of the week</i> , generalized <i>day- and nighttime</i>	<i>Frequency</i> of activity, <i>distinct days</i> of activity, <i>standard deviation</i> of activity start times	Standard deviation helps to account for the diverse start times for work-time activities.	Ahas, R et al (2010) <i>Using Mobile Positioning Data to Model Locations Meaningful to Users of Mobile Phones</i>
<i>Heuristic methods, clustering, decision rules</i>	Individualized <i>resting</i> and <i>working</i> hours	<i>Frequency</i> of activity, activity <i>location</i> (landuse)	Detecting individualized resting and working hours allows to distinguish those with unusual or variable work schedules (e.g., night shift or very early start times).	Yang, Y (2021) <i>Detecting Home and Work Locations from Mobile Phone Cellular Signaling Data</i>
<i>Heuristic and statistical methods, clustering</i>	Generalized <i>resting</i> and <i>working</i> hours, <i>weekdays</i> and <i>weekends</i>	<i>Frequency</i> of activity	Classifying the top two most-visited clusters based on two criteria: (1) high regularity and (2) temporal signature. The 'work' cluster is the highly regular location not characterized by night/weekend activity, making it flexible for non-traditional weekday schedules .	Isaacman, S et al (2011) <i>Identifying Important Places in People's Lives from Cellular Network Data</i>
<i>Heuristic methods</i>	Generalized <i>resting</i> and <i>working</i> hours, <i>weekdays</i> and <i>weekends</i>	<i>Dwell time</i> at a location	By focusing on dwell time rather than activity frequency, the method inherently handles diverse work times better by identifying the single most frequent stay during conventional work-related periods, even if the exact start time varies .	Kung, K S (2014) <i>Exploring Universal Patterns in Human Home-Work Commuting from Mobile Phone Data</i>

Considering local context

Success of home and work-time location detection can be **affected by** several **external factors** and those factors can be influenced by **local context**. Some examples include:

Seasonality¹

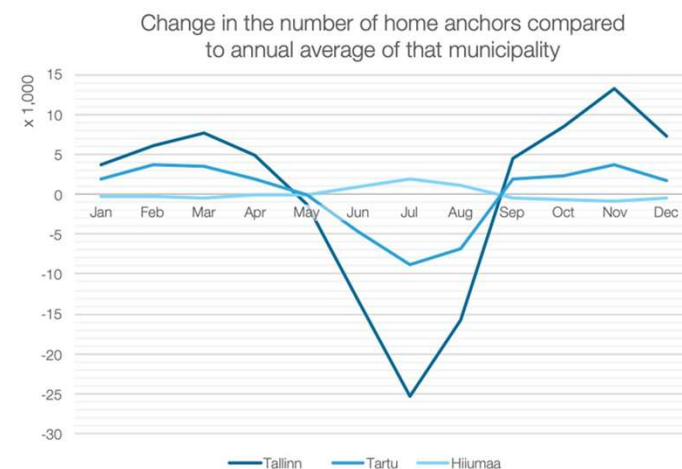
- holiday season - using shorter time frames for home detection can result in detecting holiday location as a home;
- vacations - people might not get a home and work location at all when residing abroad.

¹Silm and Ahas, R (2010) *The Seasonal Variability of Population in Estonian Municipalities*

Phone usage patterns²

- phone sharing - multiple home and work locations can be true;
- irregular phone usage - shorter time frames might not be sufficient for home and work detection;

²Warren et al (2022) *Home Location Detection from Mobile Phone Data – Evidence from Togo*

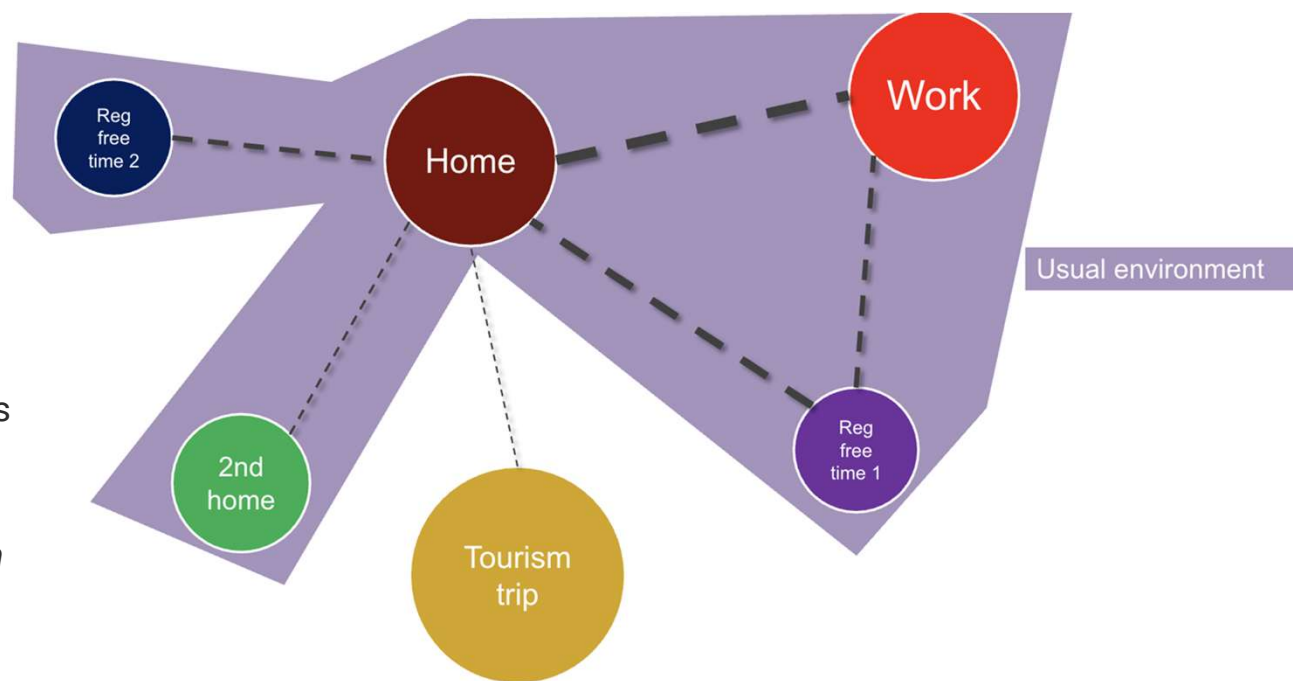


Model validation possibilities

Approach	Validation data	Main purpose	Examples
Small-scale ground truth (individual level validation)	Surveys or user addresses	Algorithm tuning and accuracy measurement	Isaacman, S et al (2011) Identifying Important Places in People's Lives from Cellular Network Data
			Söstra, K & Lehto, K (2018) Using Mobile Positioning for Improving the Quality of Register Data
			Pappalardo, L et al (2021) Evaluation of home detection algorithms on mobile phone data using individual-level ground truth
High-level comparisons (aggregate validations)	Census population and commuting data statistics	Sanity checking the results - alignment with official statistics, mobility data.	Ahas, R et al (2010) Using Mobile Positioning Data to Model Locations Meaningful to Users of Mobile Phones
			Phithakkitnukoon, S et al (2012) Socio-Geography of Human Mobility: A Study Using Longitudinal Mobile Phone Data
			Vanhoof, M et al (2018) Assessing the Quality of Home Detection from Mobile Phone Data for Official Statistics
Large-scale individual level ground truth	GPS data, labelling	Algorithm tuning and accuracy measurement	de Sojo, S et al (2025) Establishing validated standards for Home and Work location Detection (HoWDe)

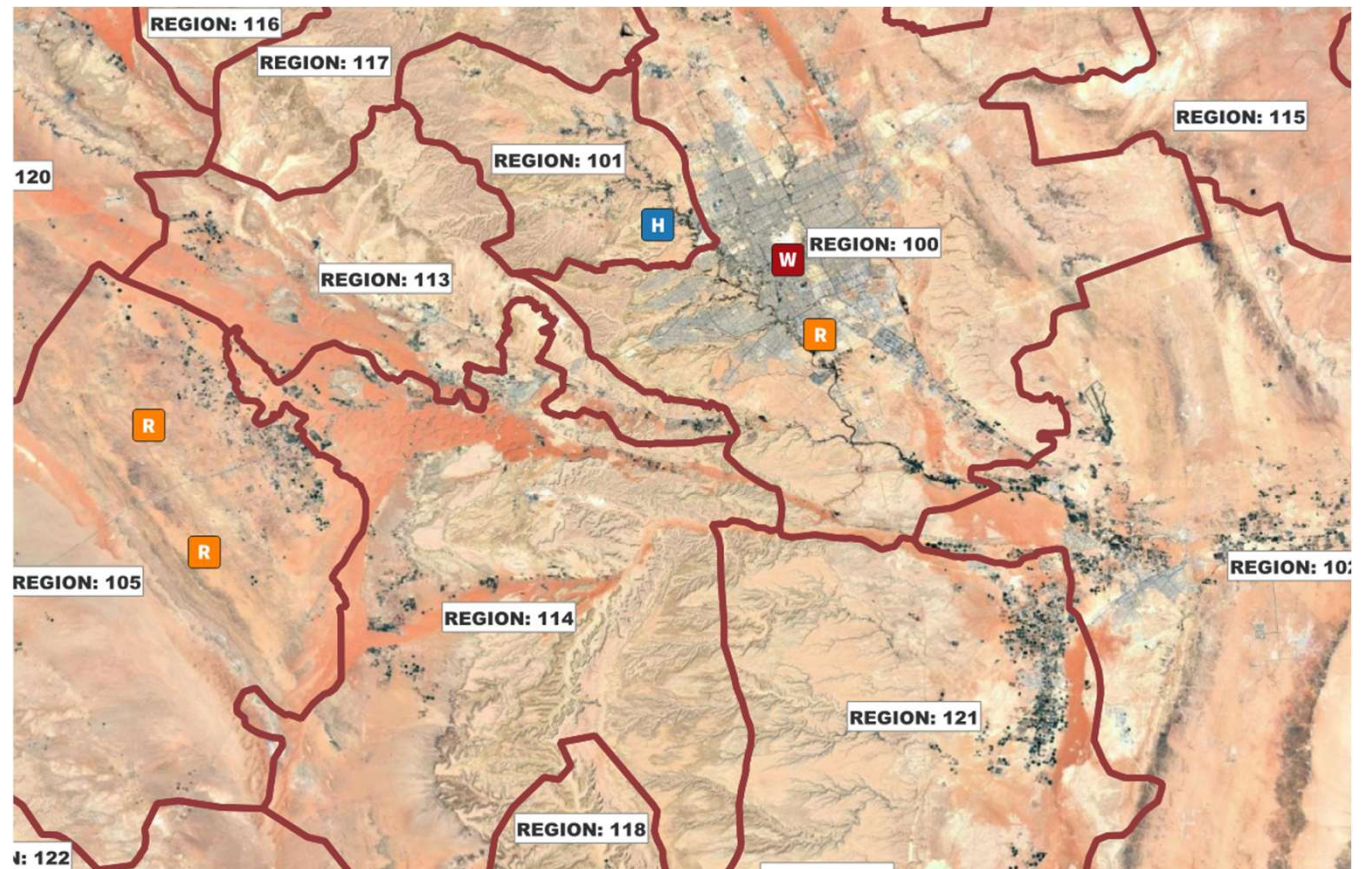
Defining usual environment with MPD

- Using MPD to determine person's usual environment (UE), **anchor points** can be used to define a geographical space person frequents to conduct their regular life routines.
- In order to define UE following needs to be defined:
 - **Anchor points** for a person;
 - **Units of geographical area** (such as LAU) that define the minimum size building blocks for UE;
 - **Criteria that defines when a person is inside UE and when not**, such as crossing the border of administrative units.



Defining usual environment with MPD

'H', 'W', 'R' are person's anchor points. This person travels from region 100 to region 105 and on the way visits also region 113. How would you define this person in region 113?



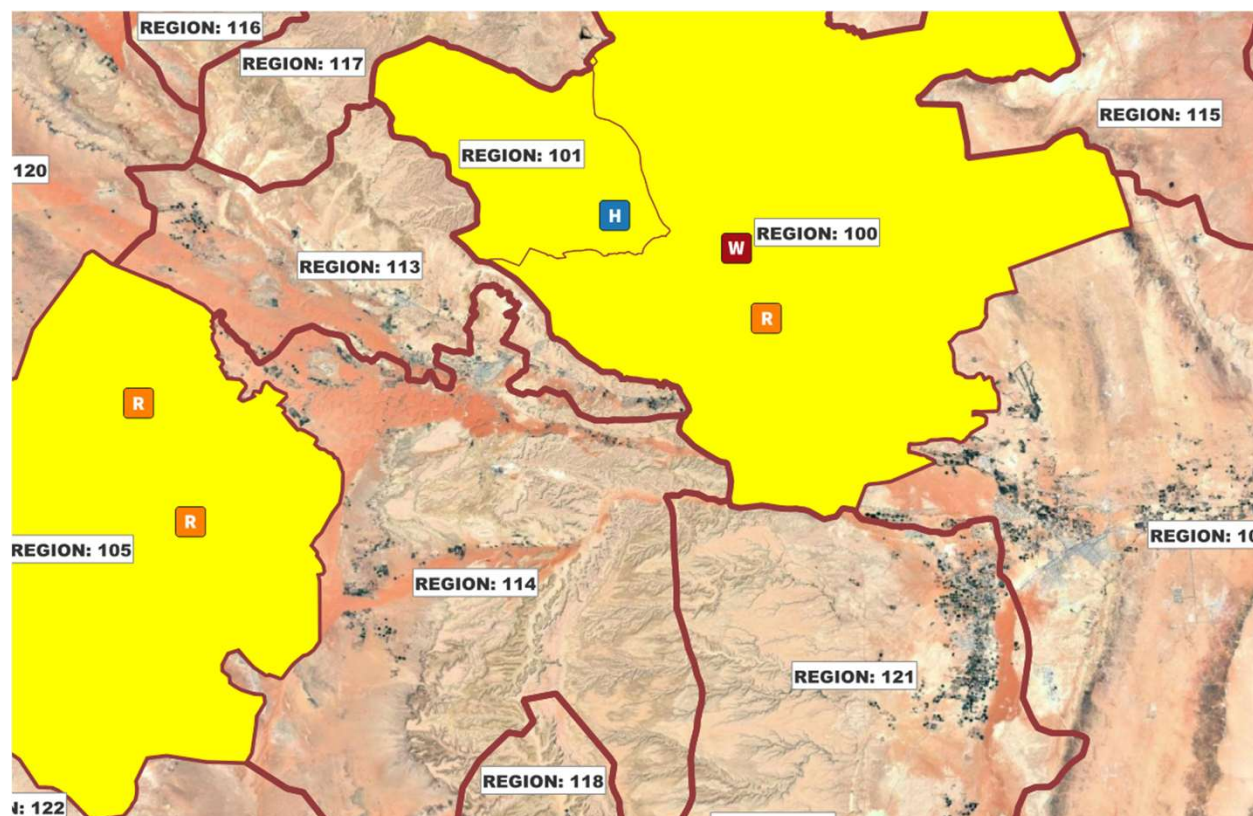
Defining usual environment with MPD

Given that **smallest unit** for UE is **region**, 100, 101, 105 should be defined as UE.

- Region 113: tourist

Note that the **answer depends** here from the **geographical unit** that is used:

- if the smallest unit for UE would be parent-region, then some of the regions contained by it and without any anchor points would be part of UE as well.

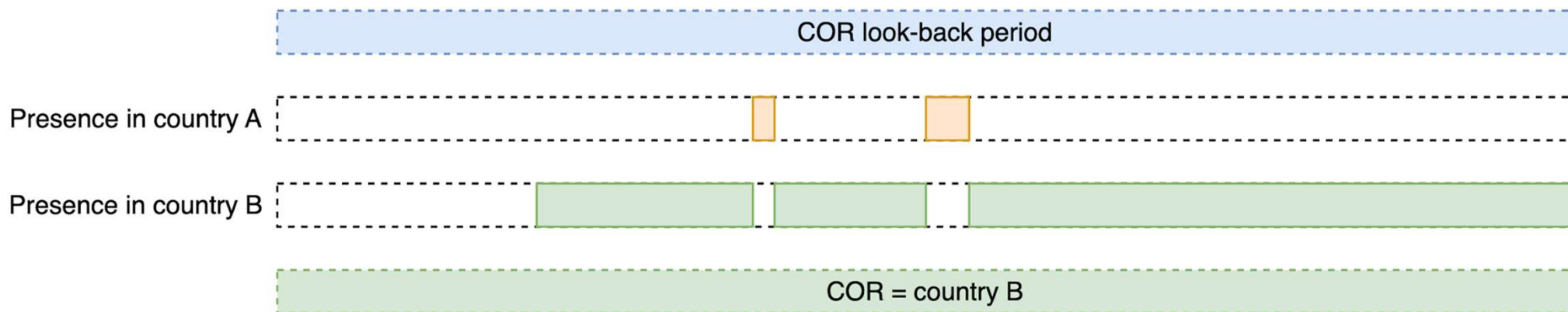


Identifying country of residence

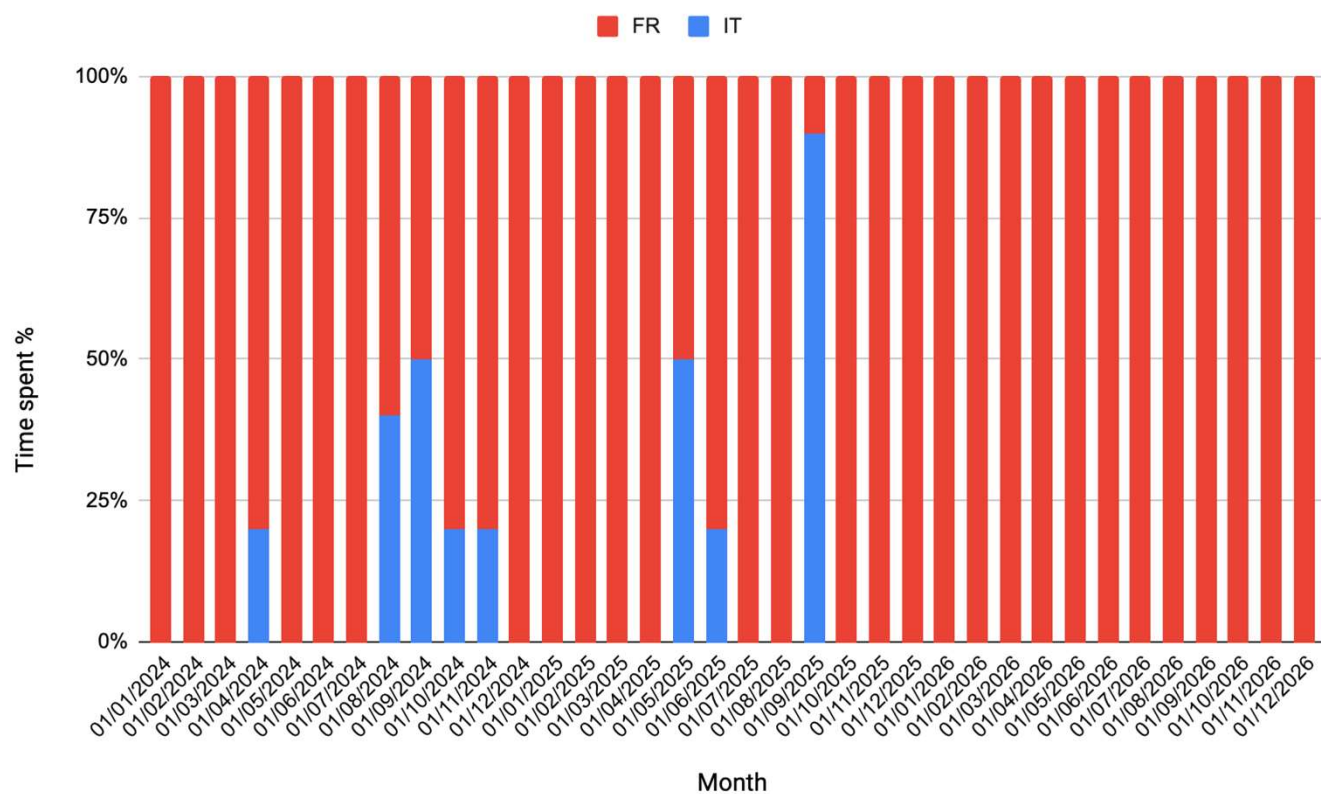
According to official definition, country of residence (COR) is the country where the person spends most of their year.

MPD perspective:

- domestic + outbound -> COR for domestic subscribers;
- inbound -> COR for foreign subscribers.

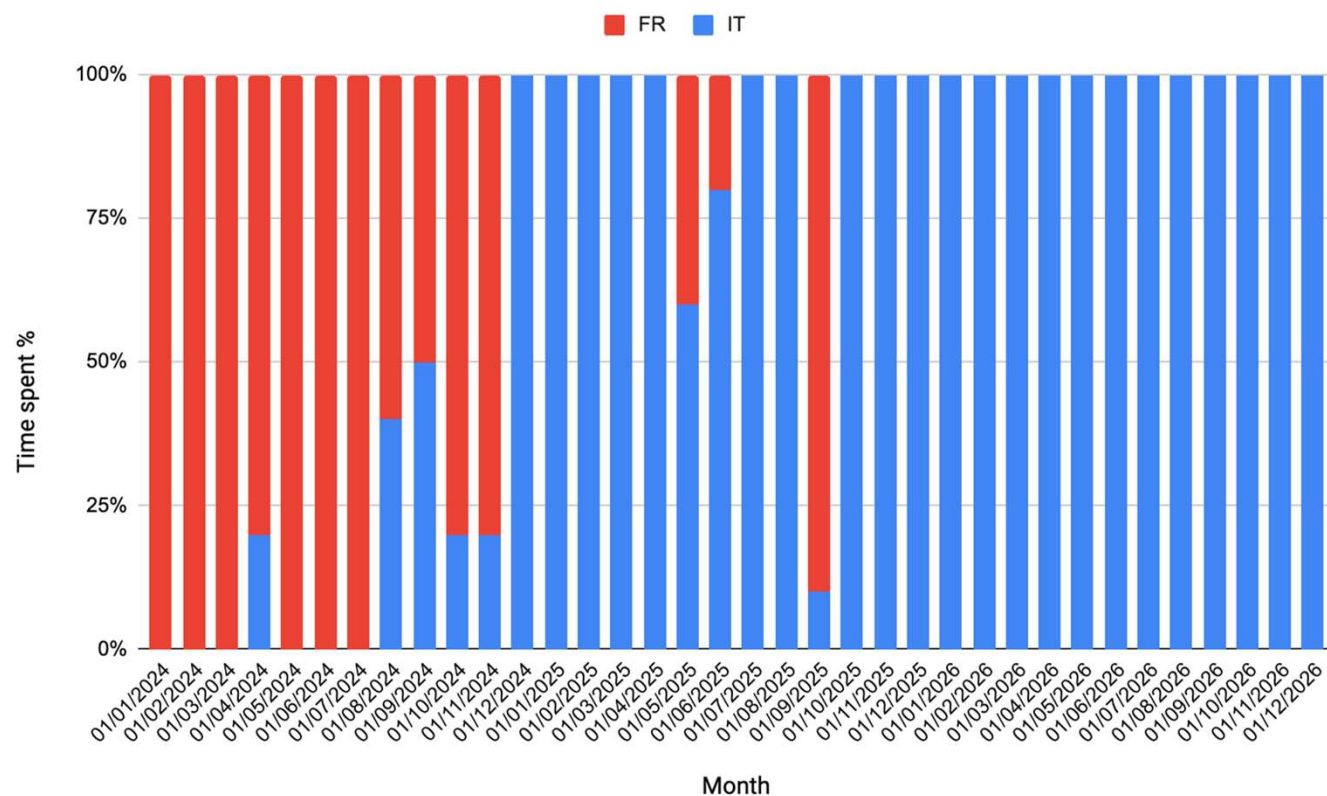


Determine country of residence



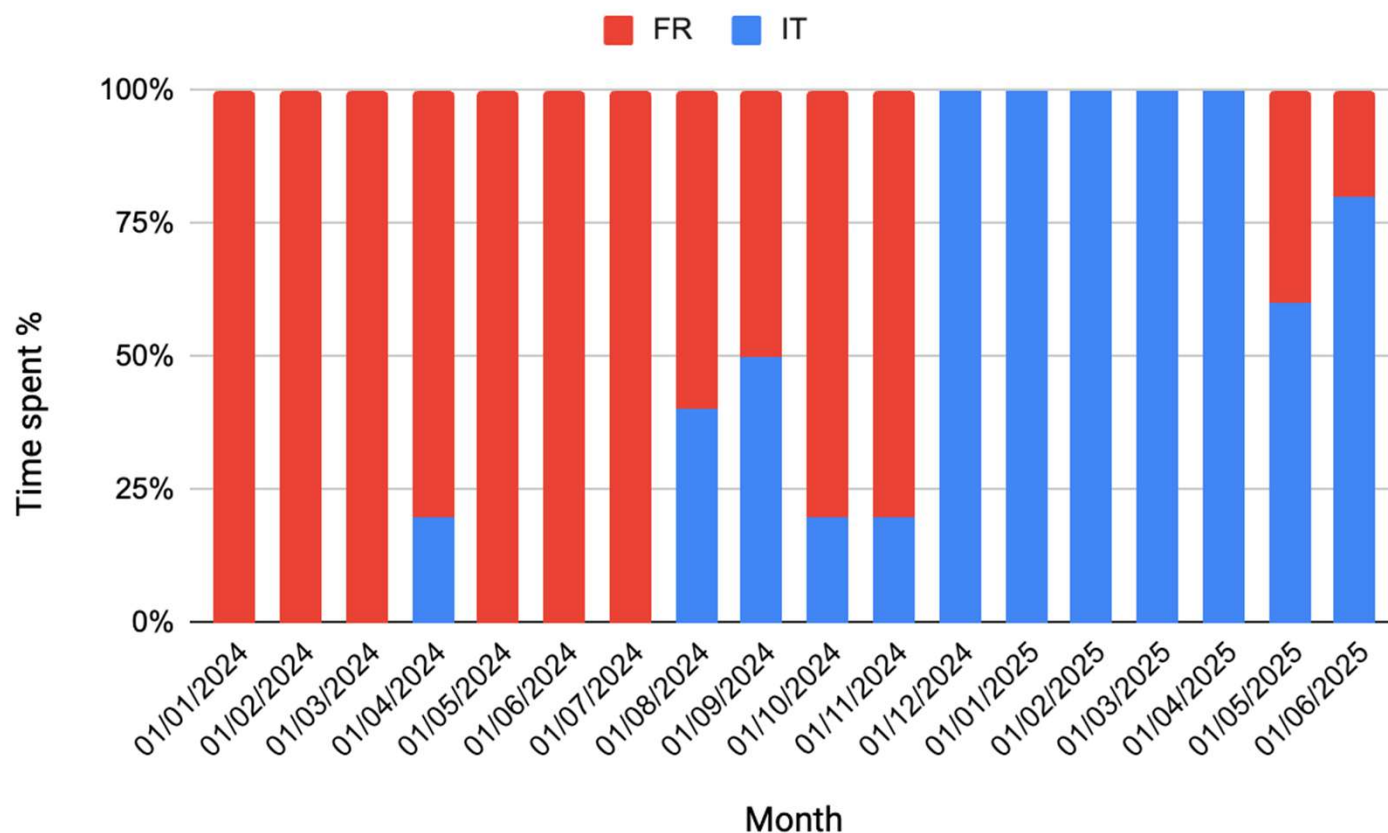
Country of residence (COR) is the country where the person spends most of their year.

Determine country of residence



Country of residence (COR) is the country where the person spends most of their year.

Determine country of residence - lookback window



Continuous updates to data require lookback window to determine COR.

Limitations and ethical considerations

In general, processing MPD for statistical purposes shall be subject to:

- **appropriate safeguards**,
- **privacy best practices**, and
- **relevant laws**, including GDPR, to protect the rights and freedoms of the data subject.

GDPR states that care should be taken to ensure that data cannot be re-identified through any reasonable effort.

Robustness evaluation for anonymization³ by EDPB:

- **singling-out** - isolating an individual in a larger group based on the data;
- **linkability** - linking together two records concerning the same individual; and
- **inference** - deducing, with significant probability, unknown information about an individual.

³EDPB (2020) Guidelines 04/2020 on the use of location data and contact tracing tools in the context of the COVID-19 outbreak.



Fundamental Principles of Official Statistics

FIVE PRINCIPLES TO ENSURE TRUST

FOLLOWING THE FUNDAMENTAL PRINCIPLES OF OFFICIAL STATISTICS

-  NECESSITY & PROPORTIONALITY
-  PROFESSIONAL INDEPENDENCE
-  PRIVACY PROTECTION
-  COMMITMENT TO QUALITY
-  INTERNATIONAL COMPARABILITY

Source: Jansen, R., Kovacs, K., Esko, S., Saluveer, E., Söstra, K., Bengtsson, L., ... Magpantay, E. (2021). Guiding principles to maintain public trust in the use of mobile operator data for policy purposes. Data & Policy, 3, E24. doi:10.1017/dap.2021.21

Introduction to Practical Exercise

MPD Continuity Models: Intro to Practical Exercise

- **Task:** Applying a speed-based continuity model to a dataset containing MPD events and cells.
- **Exercise Link:**
<https://github.com/Flowminder/WB-GDF-Modern-Data-Workflows-Code-Tasks>

The screenshot shows the GitHub repository page for 'Flowminder / WB-GDF-Modern-Data-Workflows-Code-Tasks'. The repository is public and has 20 commits. The README file is selected, showing the title 'WB-GDF: Modern Data Workflows Code Tasks' and a description: 'This is a repository containing the coding exercises for the WB-GDF Course 2: Foundations in Mobile Phone Data (MPD) for Policy: Modern Data Workflows'.

Under the 'Course Sessions' section, there is a table with the following data:

Session	Session Title	Link to Colab
Session 2.3.4	Input data quality assurance and cleaning	Open in Colab
Session 2.4.1	Continuity models	Open in Colab
Session 2.4.2	Meaningful Locations and Usual Environments	Open in Colab

A red arrow points from the exercise link in the text to the 'Session 2.4.2' row in the table.



The GDF-MPD window is a multi-partner window funded by the Spanish Government, Ministry of Economy, Commerce and Business. The GDF-MPD program is jointly managed by DEC (DECDG, DIME), Poverty and Digital Development Global Practices in technical partnership with the International Telecommunications Union (ITU).





Mobile Phone Data for Policy Programme (WB-GDF)
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